Ex. No.: 3a

Experiment Title: Shell Script – Reverse of Digit

Aim:

To write a shell script to reverse a given digit using a looping statement.

Procedure:

1. Launch the terminal and create a new shell script file named `indhu.sh`.
2. Write the script that:
   - Asks the user to enter a number.
   - Uses a `while` loop to extract and reverse the digits using modulo and division operations.
3. Save the file and provide execution permission using: `chmod +x indhu.sh`
4. Run the script with: `sh indhu.sh`
5. Check the output to confirm the number is reversed correctly.

Program:

Filename: `indhu.sh`

Unset

```
#!/bin/bash

echo "Enter a number:"
read num

rev=0
while [ $num -ne 0 ]
do
    rem=$((num % 10))
    rev=$((rev * 10 + rem))
```

```
    num=$((num / 10))
done

echo "$rev"
```

Sample Input and Output:

Unset

```
[REC@localhost ~]$ sh indhu.sh
Enter a number:
123
321
```

Result:

The shell script was executed successfully. It reversed the digits of the given number using a loop.

Ex. No.: 3b

Experiment Title: Shell Script – Fibonacci Series

Aim:
To write a shell script to generate a Fibonacci series using a `for` loop.

Procedure:

1. Open the terminal in your Linux system.
2. Create a new shell script file named `indhu.sh`.
3. Write a script that:
   - Prompts the user to enter a number.
   - Initializes the first two numbers of the Fibonacci series (0 and 1).
   - Uses a `for` loop to calculate and display the Fibonacci numbers up to the entered count.
4. Save the file and give it execute permission using `chmod +x indhu.sh`.
5. Run the script with the command: `sh indhu.sh`.
6. Observe the output and verify that the Fibonacci series is generated correctly.

Program:
Filename: `indhu.sh`

bash

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```
#!/bin/bash
```

```
echo "Enter a number"
read n

a=0
b=1

echo "Fibonacci series"
echo "$a"
echo "$b"

for (( i=2; i<n; i++ ))
do
    fn=$((a + b))
    echo "$fn"
    a=$b
    b=$fn
done
```

Sample Input and Output:

ruby

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```
[REC@localhost ~]$ sh indhu.sh
Enter number
```

```
21

Fibonacci series

0

1

1

2

3

5

8

13

21

34

55

89

144

233

377
```

Result:
The shell script was executed successfully. It generated the Fibonacci series up to the given number using a `for` loop.